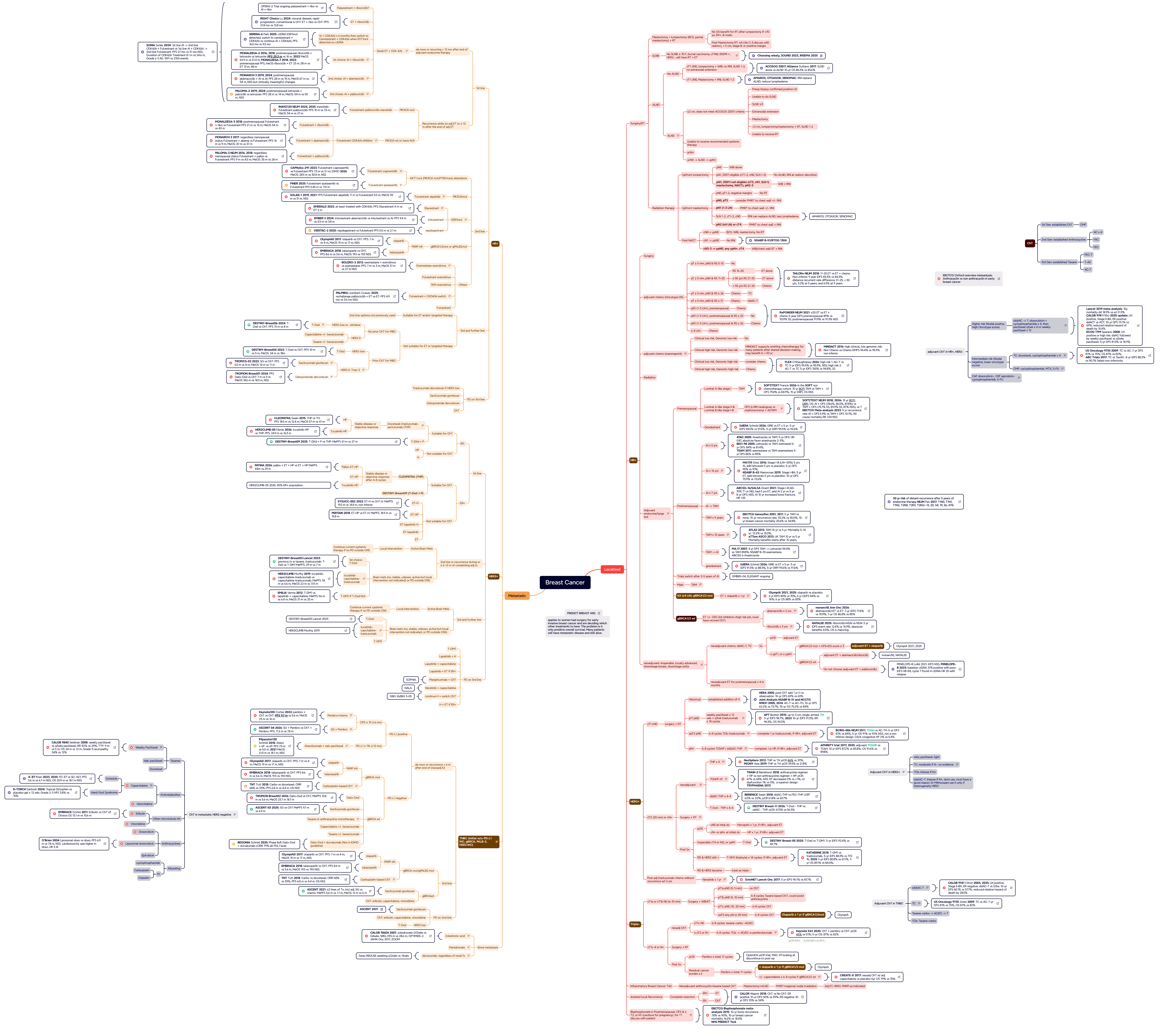

Breast Cancer

Metastatic

- **HR+**
  - 1st line
    - de novo or recurring > 12 mo after end of adjuvant endocrine therapy
      - Serial ET + CDK 4/6i
        - **SONIA** Sonke **2024**: 1st line AI -> 2nd line CDK4/6i + Fulvestrant vs 1st line AI + CDK4/6i -> 2nd line Fulvestrant: PFS 27 mo vs 31 mo NSS, Duration of CDK4/6i Treatment 8.1 m vs 24.6 m, Grade ≥ 3 AE: 1591 vs 2763 events
        - Palazestrant + ribociclib?
          - OPERA-2 Trial ongoing palazestrant + ribo vs AI + ribo
        - ET + ribociclib
          - **RIGHT Choice** Lu **2024**: visceral disease, rapid progression: conventional is ChT. ET + ribo vs ChT: PFS 21.8 mo vs 12.8 mo
        - AI + CDK4/6i x 6 months then switch to Camizestrant + CDK4/6i when EST1mut detected on ctDNA
          - **SERENA-6** Park **2025**: ctDNA ESR1mut detected, switch to camizestrant + CDK4/6i vs continue AI + CDK4/6i, PFS 16.0 mo vs 9.2 mo
        - 1st choice: AI + ribociclib
          - **MONALEESA-2 2016, 2018**: postmenopausal ribociclib + letrozole vs letrozole: ***PFS 25.3 m*** vs 16 m, **2022** MeOS 63.9 m vs 51.4 m; **MONALEESA-7 2018, 2022**: premenopausal PFS, meOS ribociclib + ET 23 m, 58 m vs ET 13 m, 48 m
        - 2nd choise: AI + abemaciclib
          - **MONARCH 3 2019, 2024**: postmenopausal abemaciclib + AI vs AI: PFS 28 m vs 15 m, MeOS 67 m vs 54 m, NSS but clinically meaningful changes
        - 3rd choise: AI + palbociclib
          - **PALOMA-2 2019, 2024:** postmenopausal letrozole + palciclib vs letrozole: PFS 28 m vs 14 m; MeOS: 54 m vs 50 m, NSS
    - Recurrence while on adj ET or ≤ 12 m after the end of adj ET
      - PIK3CA mut
        - Fulvestrant-palbociclib-inavolisib
          - **INAVO120 NEJM 2024, 2025**: inavolisib-fulvestrant-palbociclib: PFS 15 m vs 7.5 m, MeOS 34 m vs 27 m
      - PIK3CA wt or inavo N/A
        - Fulvestrant-CDK4/6 inhibitor
          - Fulvestrant + ribociclib
            - **MONALEESA-3 2018**: postmenopausal Fulvestrant + ribo vs Fulvestrant PFS 21 m vs 13 m; MeOS 54 m vs 42 m
          - Fulvestrant + abemaciclib
            - **MONARCH 2 2017**: regardless menopausal status Fulvestrant + abema vs Fulvestrant: PFS 16 m vs 9 m; MeOS: 47 m vs 37 m
          - Fulvestrant + palbociclib
            - **PALOMA-3 NEJM 2016, 2018**: regardless menopausal status Fulvestrant + palbo vs Fulvestrant: PFS 9 m vs 4.5 m; MeOS: 35 m vs 28 m
  - 2nd line
    - AKT1 mut (PIK3CA mut/PTEN loss) alterations
      - Fulvestrant-capiversitib
        - **CAPItello-291 2023**: Fulvestrant-capivasertib vs Fulvestrant PFS 7.3 m vs 3.1 m; ESMO **2026** MeOS: 28.5 m vs 30.4 m, NSS
      - Fulvestrant-ipatasertib
        - **FINER 2025**: Fulvestrant-ipatasertib vs Fulvestrant PFS 5.45 m vs 1.91 m
    - PIK3CAmut
      - Fulvestrant-alpelisib
        - **SOLAR-1 2019, 2021**: PFS Fulvestrant-alpelisib 11 m vs Fulvestrant 5.5 m; MeOS 39 m vs 31 m, NSS
    - ESR1mut
      - Elacestrant
        - **EMERALD 2022**: at least treated with CDK4/6i, PFS: Elacestrant 4 m vs ET 2 m
      - imlunestrant
        - **EMBER 3 2024**: imlunestrant-abemaciclib vs imlunestrant vs AI: PFS 9.4 m vs 5.5 m vs 3.8 m
      - vepdegestrant
        - **VERITAC-2 2025:** vepdegestrant vs fulvestrant PFS 5.0 m vs 2.1 m
    - gBRCA1/2mut or gPALB2mut
      - PARP inh
        - olaparib
          - **OlympiAD 2017**: olaparib vs ChT: PFS: 7 m vs 4 m; MeOS 19 m vs 17 m, NSS
        - talazoparib
          - **EMBRACA 2018**: talazoparib vs ChT: PFS 8.6 m vs 5.6 m, MeOS 19.3 vs 19.5 NSS
    - Others
      - Exemestane-everolimus
        - **BOLERO-2 2012**: exemestane + everolimus vs exemestane: PFS: 7 m vs 3 m; MeOS 31 m vs 27 m NSS
      - Fulvestrant-everolimus
      - TAM-everolimus
      - Fulvestrant + CKD4/6i switch
        - **PALMIRA** Llombart-Cussac **2025**: rechallenge palbociclib + ET vs ET: PFS 4.9 mo vs 3.6 mo NSS
      - Fulvestrant
  - 3rd and further line
    - Suitable for ET and/or targeted therapy
      - 2nd-line options not previously used
    - Not suitable for ET or targeted therapy
      - No prior ChT for MBC
        - HER2-low or -ultralow
          - T-Dxd
            - **DESTINY-Breast06 2024**: T-Dxd vs ChT: PFS 13 m vs 8 m
        - HER2 0
          - Capecitabine +/- bevacizumab
          - Taxane +/- bevacizumab
      - Prior ChT for MBC
        - HER2-low
          - T-Dxd
            - **DESTINY-Breast04 2022**: T-Dxd vs ChT: PFS 10 m vs 5 m; MeOS: 24 m vs 18m
        - HER2 0: Trop-2
          - Sacituzumab govitecan
            - **TROPiCS-02 2023**: SG vs ChT: PFS 5.5 m vs 4 m; MeOS 14 m vs 11 m
          - Datopotamab deruxtecan
            - **TROPION-Breast01 2024**: PFS Dato-Dxd vs ChT 7 m vs 5 m, MeOS 18.6 m vs 18.3 m, NSS
  - PD on 3rd line
    - Trastuzumab deruxtecan if HER2 low
    - Sacituzumab govitecan
    - Datopotamab deruxtecan
    - ChT
- **HER2+**
  - 1st line
    - ER-
      - Suitable for ChT
        - Docetaxel-trastuzumab-pertuzumab (THP)
          - Stable disease or objective response
            - HP
              - **CLEOPATRA** Swain **2015**: THP vs TH: PFS 18.5 m vs 12.4 m; MeOS 57 m vs 41 m
            - Tucatinib-HP
              - **HER2CLIMB-05** Dieras **2026**: tucatinib-HP vs THP: PFS: 24.9 m vs 16.3 m
        - T-DXd + P
          - **DESTINY-Breast09 2025**: T-DXd + P vs THP: MePFS 41 m vs 27 m
      - Not suitable for ChT
        - HP
        - H
    - ER+
      - Suitable for ChT
        - **CLEOPATRA (THP)**
          - Stable disease or objective response after 4-8 cycles
            - Palbo-ET-HP
              - **PATINA 2026**: palbo + ET + HP vs ET + HP MePFS 44m vs 29 m
            - ET-HP
            - Tucatinib-HP
              - HER2CLIMB-05 2026: 50% ER+ population
        - **DESTINY-Breast09 (T-Dxd + P)**
      - Not suitable for ChT
        - ET-H
          - **SYSUCC-002 2022**: ET-H vs ChT-H: MePFS: 19.2 m vs 14.8 m, non inferior
        - ET-HP
          - **PERTAIN 2018**: ET-HP vs ET-H: MePFS: 18.9 m vs 15.8 m
        - ET-lapatinib-H
        - ET-lapatinib
        - ET
  - 2nd line or recurrence during or ≤ 6-12 m of completing adj tx
    - Active Brain Mets
      - Local Intervention
        - Continue current systemic therapy if no PD outside CNS
    - Brain mets (no, stable, unkown, active but local intervention not indicated) or PD outside CNS
      - 1st choice: T-Dxd
        - **DESTINY-Breast03 Lancet 2023**: previous tx w taxane, trastuzumab: T-Dxd vs T-DM1 MePFS: 29 m vs 7 m
      - tucatinib-capcitabine-trastuzumab
        - **HER2CLIMB** Murthy **2019**: tucatinib-capecitabine-trastuzumab vs capecitabine-trastuzumab: MePFS 7.8 m vs 5.6 m, MeOS 22 m vs 17.5 m
      - T-DM1 if T-Dxd N/A
        - **EMILIA** Verma **2012**: T-DM1 vs lapatinib + capecitabine: MePFS 9.6 m vs 6.4 m; MeOS 31 m vs 25 m
  - 3rd and further line
    - Active Brain Mets
      - Local Intervention
        - Continue current systemic therapy if no PD outside CNS
    - Brain mets (no, stable, unkown, active but local intervention not indicated, or PD outside CNS)
      - T-Dxd
        - DESTINY-Breast03 Lancet 2023
      - tucatinib-capcitabine-trastuzumab
        - HER2CLIMB Murthy 2019
      - T-DM1
  - PD on 3rd line
    - T-DM1
    - Lapatinib + H
    - Lapatinib + capecitabine
    - Lapatinib + ET if ER+
    - Margetuximab + ChT
      - SOPHIA
    - Neratinib + capecitabine
      - NALA
    - continue H + switch ChT
      - GBG 26/BIG 3-05
    - H + ET if ER+
- **TNBC (initial w/u: PD-L1 IHC, gBRCA, PALB-2, HER2 IHC)**
  - de novo or recurrence > 6 m after end of (neo)adj ICI
    - PD-L1 positive
      - CPS ≥ 10 (>6 mo)
        - Pembro+chemo
          - **Keynote355** Cortes **2022:** pembro + ChT vs ChT: ***PFS 9.7 m*** vs 5.6 m, MeOS 23 m vs 16 m
        - SG + Pembro
          - **ASCENT 04 2026**: SG + Pembro vs ChT + Pembro: PFS: 11.2 m vs 7.8 m
      - PD-L1 ≥ 1% (>12 mo)
        - Atezolizumab + nab-paclitaxel
          - **IMpassion130** Schmid **2018**: Atezo + nP vs nP: PFS 7.5 m vs 5.0 m, **2021** MeOS 21.0 m vs 18.7 m, NSS
    - PD-L1 negative
      - gBRCA mut
        - PARP inh
          - olaparib
            - **OlympiAD 2017**: olaparib vs ChT: PFS: 7 m vs 4 m; MeOS 19 m vs 17 m, NSS
          - talazoparib
            - **EMBRACA 2018**: talazoparib vs ChT: PFS 8.6 m vs 5.6 m, MeOS 19.3 vs 19.5 NSS
        - Carboplatin-based ChT
          - **TNT** Tutt **2018**: Carbo vs docetaxel; ORR 68% vs 33%; PFS 6.8 m vs 4.4 m. OS NSS
      - gBRCA wt
        - Dato-Dxd
          - **TROPION Breast02 2026**: Dato-Dxd vs ChT, MePFS 10.8 m vs 5.6 m; MeOS 23.7 m 18.7 m
        - Sacituzumab govitecan
          - **ASCENT-03 2025:** SG vs ChT MePFS 9.7 m vs 6.9 m
        - Taxane or anthracycline monotherapy
        - Capecitabine +/- bevacizumab
        - Taxane +/- bevacizumab
        - Dato-Dxd + durvalumab (Not in ESMO guideline)
          - **BEGONIA** Schmid **2025**: Phase Ib/II: Dato-Dxd + durvalumab cORR 79% all PDL1 level
  - 2nd line
    - gBRCA mut/gPALB2 mut
      - PARP inh
        - olaparib
          - **OlympiAD 2017**: olaparib vs ChT: PFS: 7 m vs 4 m; MeOS 19 m vs 17 m, NSS
        - talazoparib
          - **EMBRACA 2018**: talazoparib vs ChT: PFS 8.6 m vs 5.6 m, MeOS 19.3 vs 19.5 NSS
      - Carboplatin-based ChT
        - **TNT** Tutt **2018**: Carbo vs docetaxel; ORR 68% vs 33%; PFS 6.8 m vs 4.4 m. OS NSS
    - gBRCAwt
      - Sacituzumab govitecan
        - **ASCENT 2021:** ≥2 lines of Tx, incl adj. SG vs chemo: MePFS 5.6 m vs 1.7 m; MeOS 12 m vs 6 m
      - ChT: eribulin, capecitabine, vinorelbine
  - PD on 2nd line
    - Sacituzumab govitecan
      - **ASCENT 2021**
    - ChT: eribulin, capecitabine, vinorelbine
    - HER2 low
      - T-Dxd
- Bone metastasis
  - Zoledronic acid
    - **CALGB 70604 2007**: zoledronate Q12wks vs Q4wks: SREs 29.5 m vs 28.6 m; OPTIMIZE-2 JAMA Onc 2017; ZOOM
  - Pamidronate
  - denosumab, regardless of renal fx
    - Swiss REDUSE awaiting q12wks vs 14wks

ChT in metastatic HER2 negative

- Taxanes
  - Weekly Paclitaxel
    - **CALGB 9840** Seidman **2008**: weekly paclitaxel vs q3wks paclitaxel: RR 42% vs 29%, TTP: 9 m vs 5 m; OS 24 m vs 12 m; Grade 3 neuropathy 24% vs 12%
  - Nab-paclitaxel
  - Docetaxel
- Antimetabolites
  - Capecitabine
    - Schedule
      - **X-7/7** Khan **2023, 2025**: FD-7/7 vs SD-14/7, PFS 5.6 m vs 6.7 m NSS, OS 20.9 m vs 18.1 m NSS
    - Hand-foot Syndrome
      - **D-TORCH** Santosh **2024:** Topical Diclophen vs placebo gel x 12 wks: Grade 2-3 HFS 3.8% vs 15%
  - Gemcitabine
- Other microtubule inh
  - Eribulin
    - **EMBRACE** Cortes **2011**: Eribulin vs ChT of Choice OS 13.1 m vs 10.6 m
  - Vinorelbine
- Anthracyclines
  - Doxorubicin
  - Liposomal doxorubicin
    - **O'Brien 2004** Liposomal doxo vs doxo: PFS 6.9 m vs 7.8 m, NSS; cardiotoxicity was higher in doxo, HR 3.16
  - Epirubicin
- Alkylating
  - cyclophosphamide
  - Pt
    - Carboplatin
    - Cisplatin

Localized

- Surgery/RT
  - Mastectomy = lumpectomy (BCS, partial mastectomy) + RT
    - No OS benefit for RT after lumpectomy if >70 yo, ER+, & node-
    - Post Mastectomy RT: ≥4 LNs (1-3 discuss with radonc), > 5 cm, stage III, or positive margin.
  - SLNB
    - No SLNB: ≥ 70 F, ductal carcinoma, cT1N0, ER/PR +, HER2-, will have RT + ET
      - **Choosing wisely, SOUND 2023, INSEMA 2025**
  - ALND
    - No ALND
      - cT1-2N0, lumpectomy + WBI, no RNI, SLND 1-2, no extranodal extension
        - **ACOSOG Z0011 Alliance** Guiliano **2017**: SLND alone vs ALND 10-yr OS 86.3% vs 83.6%
      - cT1-2N0, Mastectomy + RNI, SLND 1-2
        - **AMAROS, OTOASOR, SENOMAC**: RNI replace ALND, reduce lymphedema
    - ALND
      - ≤2 cm, does not meet ACOSOG Z0011 criteria
        - Preop biopsy confirmed positive LN
        - Unable to do SLND
        - SLND ≥3
        - Extranodal extension
        - Mastectomy
        - >2 cm, lumpectomy/mastectomy + RT, SLND 1-2
        - Unable to receive RT
      - Unable to receive recommended systemic therapy
      - ycN+
      - ycN0 -> SLND -> ypN+
  - Radiation therapy
    - Upfront lumpectomy
      - pN0 — WBI alone
      - pN1, Z0011 eligible (cT1-2, cN0, SLN 1-2) — No ALND; RNI at radonc discretion
      - **pN1, Z0011 not eligible (cT3, cN1, SLN 3, mastectomy, NACT); pN2-3** — WBI + RNI
    - Upfront mastectomy
      - pN0, pT1-2, negative margins — No RT
      - **pN0, pT3** — consider PMRT to chest wall +/- RNI
      - **pN1 (1-3 LN)** — PMRT to chest wall +/- RNI
      - SLN 1-2, cT1-2, cN0 — RNI can replace ALND, less lymphedema
        - AMAROS, OTOASOR, SENOMAC
      - **pN2 (≥4 LN) or cT4** — PMRT to chest wall + RNI
    - Post NACT
      - cN0-> ypN0 — BCS: WBI; mastectomy: No RT
      - cN1 -> ypN0 — No RNI
        - **NSABP B-51/RTOG 1304**
      - **cN2-3 -> ypN0, any ypN+, cT4** — WBI/chest wall RT + RNI
- **HR+**
  - Surgery
  - adjuvant chemo (Oncotype DX)
    - pT ≥ 5 mm, pN0 & RS 0-10 — No
    - pT ≥ 5 mm, pN0 & RS 11-25
      - RS 16-20 — ET alone
      - ≥ 50 y/o RS 21-25 — ET alone
      - < 50 y/o RS 21-25 — Chemo
        - **TAILORx NEJM 2018**: 11-25 ET vs ET + chemo. Non-inferior 9 year iDFS 83.3% vs 84.3%; distance recurrent rate difference 21-25, < 50 y/o, 3.2% at 5 years, and 6.5% at 9 years
    - pT ≥ 5 mm, pN0 & RS ≥ 26 — Chemo — TC
    - pT ≥ 5 mm, pN0 & RS ≥ 31 — Chemo — ddAC-T
    - pN1 (1-3 LN+), premenopausal — Chemo
    - pN1 (1-3 LN+), postmenopausal & RS ≤ 25 — No
    - pN1 (1-3 LN+), postmenopausal & RS ≥ 26 — Chemo
      - **RxPONDER NEJM 2021**: ≤25 ET vs ET + chemo 5 year iDFS premenopausal 89% vs 93.9% SS; postmenopausal 91.9% vs 91.3% NSS
    - ≥ 4 LN+ — Chemo
  - adjuvant chemo (mammaprint)
    - Clinical low risk, Genomic low risk — No
    - Clinical high risk, Genomic low risk — MINDACT supports omitting chemotherapy for many patients after shared decision-making, may benefit in < 50 yr
      - **MINDACT 2016**: high clinical, low genomic risk: Non Chemo vs Chemo DMFS 94.4% vs 95.9% non inferior
    - Clinical low risk, Genomic high risk — consider chemo
    - Clinical high risk, Genomic high risk — Chemo
      - **FLEX** O'Shaughnessy **2026**: high risk 1: AC-T vs TC 3-yr iDFS 95.9% vs 95.9%, NSS; high risk 2: AC-T vs TC 3-yr iDFS 100% vs 94.8%, SS
  - Radiation
  - Adjuvant endocrine/targeted
    - Premenopausal
      - Luminal A-like stage I — TAM
        - **SOFT/TEXT** Francis **2026**: In the **SOFT** non chemotherapy cohort: 15 yr BCFI TAM vs TAM + OFS 79.4% vs 84.9%; 15 yr DRFI, OS NSS
      - Luminal A-like stage II-III, Luminal B-like stage I-III — OFS (LHRH analogues vs oophorectomy) + AI/TAM
        - **SOFT/TEXT NEJM 2018, 2026:** 15 yr BCFI, DRFI, OS: AI + OFS (78.6%, 86.5%, 87.8%) vs TAM + OFS (75.7% SS, 83.9% SS, 87% NSS); vs T **EBCTCG Meta-analysis 2022**: 5 yr recurrence rate AI + OFS 6.9% vs TAM + OFS 10.1%; All cause mortality RR 1.04 NSS
      - Giredestrant
        - **lidERA** Schmid **2026**: GIRE vs ET x 5 yr: 3-yr iDFS 94.0% vs 91.5%; 3-yr DRFI 95.5% vs 92.6%
    - Postmenopausal
      - AI x 5 yrs
        - **ATAC 2005**: Anastrazole vs TAM: 5-yr DFS: HR 0.87, absolute favor anastrazole 2-3%; **BIG1-98 2005**: Letrozole vs TAM: estimated 5-yr DFS: 84% vs 81.4%; **TEAM 2011**: exemestane vs TAM-exemestane 5-yr DFS 86% vs 85%
      - AI x 10 yrs
        - **MA17.R** Goss **2016**: Stage I-III (LN+ 50%) 5 yrs AI, add letrozole 5 yrs vs placebo: 5-yr DFS 95% vs 91%; **NSABP B-42** Mamounas **2019**: Stage I-IIIA: 5 yr ET, add letrozole 5 yrs vs placebo: 10-yr DFS 75.9% vs 72.6%
      - AI x 7 yrs
        - **ABCSG-16/SALSA** Gnant **2021**: Stage I-III (60-70% T1 or N0), had 5 yrs ET, add AI 2 yr vs 5 yr: 8-yr DFS: NSS; AI 10 yr increased bone fracture, HR 1.35
      - AI -> TAM
      - TAM x 5 years
        - **EBCTCG tamoxifen 2001, 2011**: 5 yr TAM vs none: 15-yr recurrence rate: 33.2% vs 45.0%; 15-yr breast cancer mortality 25.6% vs 34.8%
      - TAM x 10 years
        - **ATLAS 2013**: TAM 10 yr vs 5 yr: Mortality 5-14 yr: 12.2% vs 15.0%; **aTTom ASCO 2013**: UK TAM 10 yr vs 5 yr Mortality benefits starts after 10 years;
      - TAM -> AI
        - **MA.17 2007:** 4 yr DFS TAM -> Letrozole 94.4% vs TAM 89.8%; NSABP B-33 exemestane; ABCSG 6 Anastrozole
      - giredestrant
        - **lidERA** Schmid **2026**: GIRE vs ET x 5 yr: 3-yr iDFS 91.3% vs 88.3%; 3-yr DRFI 93.6% vs 91.6%
  - Trials switch after 2-5 years of AI
    - EMBER-04, ELEGANT ongoing
  - Male — TAM
  - **N2 (≥4 LN): gBRCA1/2 mut** — ET + olaparib x 1 yr
    - **OlympiA 2021, 2025**: olaparib vs placebo: 6 yr IDFS 80% vs 70%; 6 yr DDFS 84% vs 76%; 6 yr OS 88% vs 83%
  - **gBRCA1/2 wt** — ET +/- CKD 4/6 inhibitors (high risk pts, must have recived ChT)
    - abemaciclib x 2 yrs
      - **monarchE Ann Onc 2026**: abemaciclib+ET vs ET: 7-yr iDFS 77.4% vs 70.9%; 7-yr OS 86.8% vs 85%
    - ribociclib x 3 yrs
      - **NATALEE 2025:** ribociclib+NSAI vs NSAI: 5 yr iDFS event rate 12.4% vs 15.9%, absolute benefits 4.5%; OS is maturing
  - neoadjuvant: inoperable, locally advanced, downstage breast, downstage axilla
    - neoadjuvant chemo: ddAC-T, TC — Sx
      - pCR — adjuvant ET
      - ≥ ypT1, or ≥ ypN1
        - gBRCA1/2 mut + CPS+EG score ≥ 3 — **adjuvant ET + olaparib**
          - OlympiA 2021, 2025
        - gBRCA1/2 wt
          - adjuvant ET + abemaciclib/ribociclib
            - monarchE, NATALEE
          - Do not choose (adjuvant ET + palbociclib)
            - PENELOPE-B Loibl 2021: iDFS NSS; **PENELOPE-B 2023:** baseline ctDNA 7/78 positive with poor iDFS HR 8.8, cycle 7 found 4 ctDNA HR 25 with relapse
    - neoadjuvant ET for postmenopausal x 4-6 months
- **HER2+**
  - cT1 cN0 — surgery + RT
    - Historical — established addition of H
      - **HERA 2005:** post ChT add 1 yr H vs observation: 10-yr DFS 69% vs 63% **Joint Analysis NSABP B-31 and NCCTG N9831 2005, 2014**: AC-T vs AC-TH, 10-yr DFS 62.2% vs 73.7%, 10-yr OS 75.2% vs 84%
    - pT1 pN0 — weekly paclitaxel x 12 wks + q3wk trastuzumab x 18 cycles
      - **APT** Boston **2015:** up to 3 cm, single-armed. TH 3-yr IDFS 98.7%; **2023** 10-yr iDFS 91.3%, RFI 96.3%, OS 94.3%
    - ≥pT2 pN0 — 6-8 cycles TCb-trastuzumab — complete 1 yr trastuzumab, if HR+, adjuvant ET
      - **BCIRG-006 NEJM 2011:** TCbH vs AC-TH: 5-yr DFS 81% vs 84%; 5-yr OS 91% vs 92% NSS; not a non inferior design. G3/4 congestive HF 2% vs 0.4%
    - pN+ — 6-8 cycles TCbHP / (dd)AC-THP — complete 1 yr HP, if HR+, adjuvant ET
      - **APHINITY trial 2017, 2025:** adjuvant TCbHP vs TCbH: 10 yr IDFS 87.2% vs 83.8%; OS 91.6% vs 89.8%
  - cT2 (20 mm) or cN+
    - neoadjuvant
      - THP x 4
        - **NeoSphere 2012:** THP vs TH: pCR 46% vs 29%; **PEONY** Asia **2019**: THP vs TH: pCR 39.3% vs 21.8%
      - TCbHP x9
        - **TRAIN-2** Ramshorst **2018**: anthracycline regimen + HP vs non anthracycline regimen + HP: pCR: 67% vs 68%, NSS. EF decreased 2% vs <1%; LV dysfunction 1% vs 0%; a superior design **TRYPHAENA 2013**
      - ddAC-THP x 4-4
        - **BERENICE** Swain **2018**: ddAC-THP vs FEC-THP. LVEF 6.5% vs 2.0%; pCR 61.8% vs 60.7%
      - T-Dxd - THP x 4-4
        - **DESTINY Breast-11 2026**: T-Dxd - THP vs ddAC - THP: pCR: 67.3% vs 56.3%
    - Surgery + RT
    - Post Sx
      - pCR
        - cN0 at intial dx — Herceptin x 1 yr, if HR+, adjuvant ET
        - cN+ or pN+ at initial dx — HP x 1 yr, if HR+, adjuvant ET
      - inoperable (T4 or N2), or ypN1 — T-Dxd
        - **DESTINY Breast-05 2025:** T-Dxd vs T-DM1: 3-yr iDFS 92.4% vs 83.7%
      - RD & HER2 still + — T-DM1 (Kadcyla) x 14 cycles, if HR+, adjuvant ET
        - **KATHERINE 2018**: T-DM1 vs trastuzumab, 3-yr iDFS 88.3% vs 77.0 %; **2025** 7-yr iDFS 80.8% vs 67.1%, 7-yr OS 89.1% vs 84.4%;
      - RD & HER2 became - — treat as triple -
  - Post adj trastuzumab-chemo without recurrence w/i 2 yrs — Neratinib x 1 yr
    - **ExteNET Lancet Onc 2017**: 5 yr iDFS 90.1% vs 87.7%
- **Triple-**
  - cT1a or cT1b N0 (6-10 mm) — Surgery + WBrRT
    - pT1a pN0 (0, 5 mm] — no ChT
    - pT1b pN0 (5, 10 mm] — 6-8 cycles Taxane based ChT, could avoid anthracycline
    - pT1c pN0 (10, 20 mm] — 6-8 cycles ChT
    - ≥pT2 any pN (≥ 20 mm) — 6-8 cycles ChT — **Olaparib x 1 yr if gBRCA1/2mut**
      - OlympiA
  - cT1c-4 or N+
    - neoadj ChT
      - cT1c N0 — 6-8 cycles: taxane-carbo -AC/EC
      - ≥ cT2 or N+ — 6-8 cycles: TCb -> AC/EC w pembrolizumab
        - **Keynote 522 2020**: ChT + pembro vs ChT: pCR: 65% vs 51%; 5-yr OS: 87% vs 82%
        - pCR 65%
        - 5-OS 82% to 87%
    - Surgery + RT
    - Post Sx
      - pCR — Pembro x total 17 cycles
        - OptimICE-pCR trial, MAC 29 looking at discontinue ici post op
      - Residual cancer burden ≥ II — Pembro x total 17 cycles
        - **+ olaparib x 1 yr if gBRCA1/2 mut**
          - OlympiA
        - +/- capecitabine x 6-8 cycles if gBRCA1/2 wt
          - **CREATE-X 2017:** neoadj ChT w/ adj capecitabine vs placebo 5yr OS 79% vs 70%
- Inflammatory Breast Cancer T4d — Neoadjuvant anthrocyclin+taxane based ChT — Mastectomy+ALND — PMRT+regional nodal irradiation — Adj ET, HER2, PARPi as indicated
- Isolated local Recurrence — Complete resection
  - ER+ — ET
  - ER- — ChT
    - **CALOR** Wapnir **2018**: ChT vs No ChT: ER positive 10-yr DFS 50% vs 59%; ER negative 10-yr DFS 70% vs 34%
- Bisphosphonate in Postmenopausal, OFS & ≥ T2, or N1 (cautious for pregnancy), for T1 discuss with patient
  - **EBCTCG Bisphosphonate meta-analysis 2015**: 10-yr bone recurrence 7.8% vs 9.0%, 10-yr breast cancer mortality 16.6% vs 18.4% **NHS PREDICT Tool**

PREDICT BREAST HNS

applies to women had surgery for early invasive breast cancer and are deciding which other treatments to have. The problem is it only predicts overall survival. Many patients will have metastatic disease and still alive.

ChT

- 1st Gen: established ChT — CMF
- 2nd Gen: established Anthracycline
  - AC x 4
  - FAC
  - FEC
- 3rd Gen: established Taxane
  - FEC-T
  - T-AC
  - AC-T

EBCTCG Oxford overview metaanlysis: Anthracyclin vs non anthracyclin in early breast cancer

adjuvant ChT in HR+, HER2-

- Higher risk (Nodal positve, High Oncotype score)
  - (dd)AC -> T: doxorubicin + cyclophsophamide x 4, then paclitaxel q2wk x 4 or weekly paclitaxel x 12
    - **Lancet 2019 meta-analysis**: 10y mortality dd 18.9% vs sd 21.3% **CALGB 9741** Filho **2025 update:** LN positive, Stage II-IIIA, ER positive: ddACT vs ACT, 10-yr DFS 73.1% vs 69%; reduced relative hazard of death by 15.4% **ECOG 1199** Sparano **2008**: LN positive or high risk: ddAC followed by weekly paclitaxel vs q3wks paclitaxel: 5-yr DFS 81.5% vs 76.9%
- Intermediate risk (Nodal negative, lower oncotype score)
  - TC: docetaxel, cyclophosphamide x 4
    - **US Oncology 9735 2009**: TC vs AC: 7-yr DFS 81% vs 75%, OS 87% vs 82%; **ABC Trials 2017:** TC vs TaxAC: 4-yr iDFS 88.2% vs 90.7% failed non-inferiority
  - CMF: cyclophosphamide, MTX, 5-FU
- CAF doxorubicin-, CEF epirubicin-cyclophosphamide, 5-FU

20 yr risk of distant recurrence after 5 years of endocrine therapy **NEJM** Pan **2017**: T1N0, T1N1, T1N2, T2N0, T2N1, T2N2= 13, 20, 34, 19, 26, 41%

Adjuvant ChT in HER2+

- wkly paclitaxel: light
- TC: moderate if N-, no evidence
- TCb: Intense if N+
- (dd)AC-T: Intense if N+, dont use, must have a good reason. Dr Mithoowani use it only if heterogeneity HER2

Adjuvant ChT in TNBC

- (dd)AC-T
  - **CALGB 9741** Citron **2003, 2025**: LN positive, Stage II-IIIA, ER negative: ddAC-T vs Q3w: 10-yr DFS 68.1% vs 57.7%, reduced relative hazard of death by 28.5%
- TC
  - **US Oncology 9735** Jones **2009**: TC vs AC: 7-yr DFS 81% vs 75%, OS 87% vs 82%
- Taxane-carbo -> AC/EC -> T
- TCb: Taxane-carbo

Presented with **xmind**